DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2025-26
MONDAY TEST

Class: X **Total Marks: 40**
Subject: Robotics & AI **Date: 16/05/25**

Instruction:
Attempt all questions. The intended marks for questions are given in brackets [ ].
Write variable description table for all programs. This paper consists of three printed pages.

Question 1 **[1x10=10]**

i) Smart elevators can improve energy efficiency by:
a) Running slower b) Turning off floor lights
c) Optimizing trips based on usage patterns d) Avoiding maintenance

ii) Which of the following is **NOT** typically a feature of assistant robots?
a) Voice recognition b) Facial expression detection
c) Rocket propulsion d) Touch sensors

iii) Which technology do warehouse robots often use to navigate within a warehouse?
a) Radar b) LiDAR c) Sonar d) Bluetooth

iv) Which of the following best describes a list?
a) Ordered collection of items b)Unordered collection of items
c) Collection of key-value pairs d) Immutable sequence

v) What will be the result of [1, 2] * 2**?**
a) [1, 2, 1, 2] b) [2, 4] c) [1, 1, 2, 2] d) [1, 2, 2, 4]

vi) What is the **output** of list("ABC")**?**
a) ['ABC'] b) ['A', 'B', 'C'] c) [ABC] d) ('A', 'B', 'C')

vii) What will **mylist[1:3] return** if mylist = [10, 20, 30, 40, 50]**?**
a) [10, 20, 30] b) [20, 30] c) [30, 40] d) [20, 30, 40]

viii) Which of the following methods are valid for a **tuple?**
a) append() b) insert() c) count() d) remove()

ix) If **t = (1, 2, 3, 2, 4),** what will **t.index(2)** return?
a) 1 b) 2 c) 3 d) 4

x) What is a key feature of a deterministic system?
a) Its outputs are random
b) Its behavior changes with emotions
c) It always produces the same output for the same input
d) It requires human supervision

Question 2

i) Differentiate between conventional programming approach and machine learning programming approach with the help of the following example.
Calculate speed by taking distance and time. **[4]**

ii) Consider the following list myList. What will be the elements of myList after the following 2 operations? **[2]**
myList = [10,20,30,40]
(a) print(myList.append([50,60])) (b) print(myList.extend([80,90]))

iii) *"Arya and Rohan are both learning how to ride bicycles. Arya uses a smart helmet that provides real-time alerts whenever she shifts her balance incorrectly or slows down too much, helping her adjust as she rides. Rohan, on the other hand, prefers exploring different terrains without any feedback tools, relying entirely on his instincts and trial-and-error to figure out what works."*
If a machine learning model were trained the way Arya and Rohan are learning, which type of learning (supervised or unsupervised) would each represent? Explain how the presence or the absence of structured feedback plays a role in your reasoning. **[2]**

iv) Write **any two** benefits of healthcare robots. **[2]**

Question 3

i) Write a python program that performs the following operations on a list of fruits: **[8]**
(a) Create a list of fruits ['apple','banana','cherry','mango']
(b) Insert the fruit **'grape'** at index **2**
(c) Find the index of **'mango'** and print it.
(d) Sort the list alphabetically in **descending** order.
(e) Remove **'cherry'** from the list.
(f) **Convert** the final list into a tuple.
(g) Print the **tuple** and demonstrate accessing the **third element** of the tuple.
(h) Delete the **original list** created.

ii) Explain how autonomous drones are helpful in search and rescue operations**.** **[2]**

Question 4

i) Write a Python program to create the record of a student as a list named **stRecord** with name of the student as '**Raman'**, RollNo of the student as '**A-36'**, Marks in five subjects in a nested list **[56,98,99,72,69]**. **[8]**

(a) Calculate the percentage of the student when the total marks are 100 in each subject and append it to the above list.
(b) Print the marks in the **fifth** subject from the list.
(c) Print the **maximum marks** of the student from the list.
(d) Change the name of the Student from **'Raman'** to **'Raghav'** and print the name from the list.

ii) How does decision making by humans differ from that of the machines? Explain with an example. **[2]**